RIVET-A3 Multi-Key, Multi-Context AES Cryptographic Subsystem

Technical Design and Verification Report

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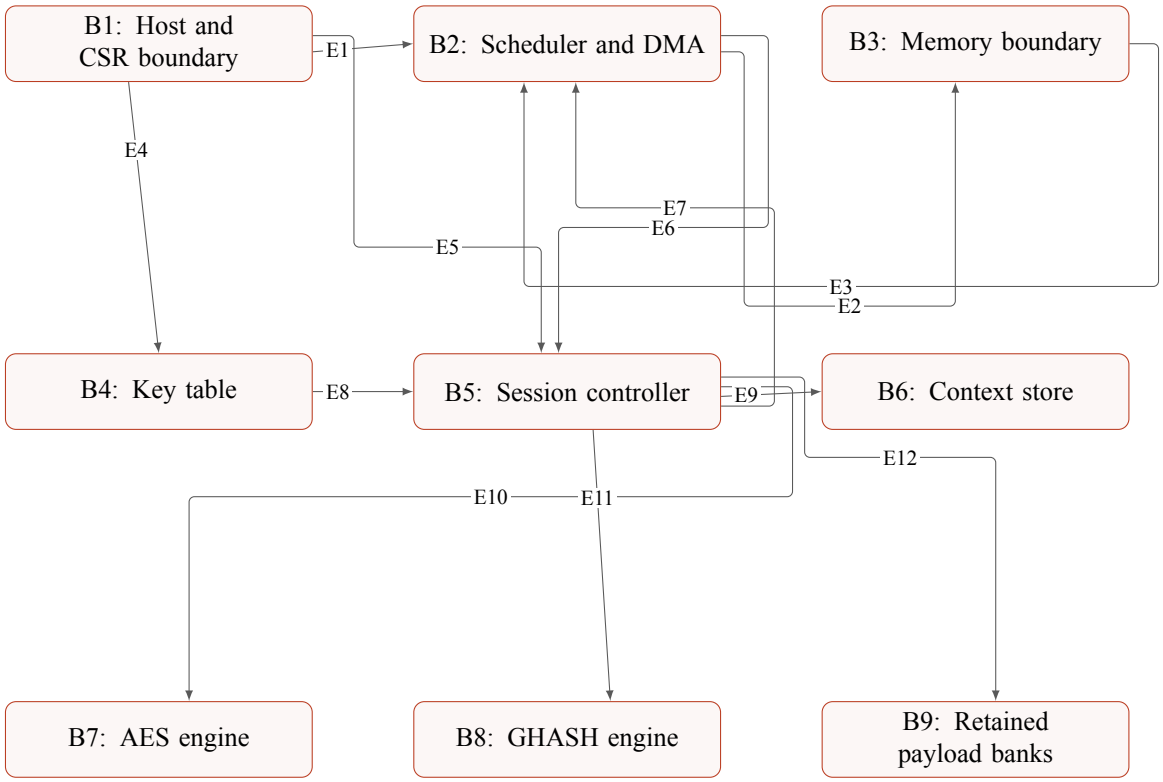
Abstract

RIVET-A3 is a standalone SystemVerilog cryptographic subsystem supporting AES-128/256 ECB, CTR and GCM through legacy streaming, direct descriptors and descriptor DMA. Four hardware keyslots and four saved contexts enable interleaved GCM sessions without software handling intermediate cryptographic state. Authentication-gated payload retention prevents failed GCM decryption from releasing plaintext.

The behavioral campaign passed all nine combinations of capacities 1, 3 and 16 blocks with seeds 17, 89 and 203. Additional checks exercised the 256-block maximum, rejected illegal capacities, behaviorally detected a disposable cryptographic mutation and reproduced a focused campaign from copied deliverable sources. Default-capacity generic synthesis reported 2,200,117 generic primitive cells and zero inferred latches. These results establish bounded RTL functionality and generic synthesizability, not physical implementation or security certification.

1 Architecture and design rationale

A shared iterative AES engine and a 128-cycle polynomial GHASH multiplier serve a byte-serial session controller. Sharing avoids four replicated cryptographic engines while preserving arbitrary-byte GCM fragment boundaries. Four saved records preserve continuation state; separate payload banks retain output and destination ownership across descriptors. The scheduler owns DMA queues, descriptor fetching, memory transactions and completion delivery. Fully resettable storage favors explicit zeroization over technology-specific RAM inference. Blocks are arranged row-major in three columns; relationships represent concurrent control, data and ownership paths, not a serial pipeline. Architecture and implementation are documented by E001, E002 and E014.



Functional block map, not a gate-level netlist. Edge identifiers match the directed relationships below. Crossings do not denote junctions; separate attachment points preserve arrow directions.

Block	Responsibility
B1: Host and CSR boundary	Legacy commands, streaming data, direct descriptors, DMA submissions, key operations and status registers.
B2: Scheduler and DMA	u_scheduler_dma implements independent depth-two high/normal FIFOs, descript or dispatch, aligned memory requests, status writes and retained completions.
B3: Memory boundary	External 128-bit request/response interface with 32-bit byte addresses and exactly one outstanding accepted read or write.
B4: Key table	u_key_table stores four normalized keys with independent length, validity and reset-lifetime lock; bound or locked replacement is rejected.
B5: Session controller	Validates descriptors, restores contexts, consumes bytes, sequences cryptography, finalizes results and scrubs consumed state.
B6: Context store	u_context_store contains four 1280-bit records with combinational reset-gated reads and edge-committed clear/write operations.
B7: AES engine	u_aes_core contains adapted key-schedule, datapath, S-box, ShiftRows and MixColumns modules; supplies ECB transforms, keystream, H and tag masks.
B8: GHASH engine	u_ghash performs polynomial multiplication for ciphertext authentication and aggregate-length processing.
B9: Retained payload banks	Four context-owned byte banks and per-byte destination maps retain enhanced output until finalization and retirement.

Directed relationship	Meaning
E1: B1 to B2	Descriptors, submissions and completion handshake
E2: B2 to B3	Memory requests
E3: B3 to B2	Memory responses and faults
E4: B1 to B4	Atomic key loads and result consumption
E5: B1 to B5	Legacy commands, streams and CSR control
E6: B2 to B5	Descriptor dispatch and status-fault notification

Directed relationship	Meaning
E7: B5 to B2	Byte I/O, results and scrub acknowledgement
E8: B4 to B5	Selected key, length and validity
E9: B5 to B6	Context selection, commit and clear
E10: B5 to B7	Key/block sequencing and AES result consumption
E11: B5 to B8	Hash operands and product consumption
E12: B5 to B9	Retain, authenticated drain and scrub

2 Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single clock and asynchronous active-low reset. Public outputs are zero during reset; availability begins after the first release rising edge.
Key channel	Input/output	256-bit key; 2-bit slot; 4-bit result	key_valid/key_ready accepts key_in, key_len_sel, key_slot_id and key_lock. Length select 0 uses key_in[255:128] for AES-128 and clears the unused half; 1 selects AES-256. key_result_valid/code remain until key_result_ready. key_loaded reflects slot0 validity.
Legacy raw ECB	Input/output	128-bit block; 8-bit transaction ID	in_valid/in_ready transfers in_block, in_enc_dec and in_tag_id using slot0. Shared output echoes ID and direction. Unloaded attempts assert err_unloaded_key without accepting work.
Legacy GCM command and streaming input	Input/output	96-bit IV; two 32-bit lengths; 128-bit tag/data; 5-bit byte count	cmd_valid/cmd_ready accepts cmd_decrypt, iv_in, aad_bytes, payload_bytes and expected_tag. data_valid/data_ready transfers data_is_aad, data_in and valid_bytes, AAD before payload. Each beat supplies exactly min(16, remaining section bytes) MSB-valid bytes.
Shared payload and legacy tag outputs	Output with input readiness	128-bit payload/tag; 8-bit ID; 5-bit byte count	out_valid/out_ready carries out_block, out_tag_id, out_enc_dec, out_valid_bytes and out_is_gcm. Unused bytes are zero; enhanced output ID is zero. tag_valid/tag_ready carries tag_out; tag_match, auth_fail and protocol_error describe the legacy/shared result. Legacy GCM retires its tag before payload.
Direct task descriptor	Input with output readiness	512 bits	task_valid/task_ready accepts task_descriptor. Direct jobs ignore descriptor addresses and use streaming input; enhanced and legacy input ownership cannot mix.
DMA submission and completion	Input/output	32-bit descriptor address; 8-bit cookie; 2-bit context; 4-bit code; 128-bit tag	dma_submit_valid/ready accepts dma_desc_addr, dma_priority and dma_cookie; priority 1 selects high. completion_valid/ready retains completion_cookie, completion_ctx_id, completion_code, completion_final and completion_tag. DMA uses the submission cookie; direct work uses descriptor byte56.
Memory request/response	Output requests; input responses	32-bit address; 128-bit data; 16-bit write strobes	mem_req_valid/ready transfers mem_req_write, mem_req_addr, mem_req_wdata and mem_req_wstrb. mem_rsp_valid/ready transfers mem_rsp_rdata and mem_rsp_error. Requests are 16-byte aligned; lane i maps address+i to bits[8*i+:8]. Reads have zero data/strobes. Every accepted request, including writes, requires one response before another request.

Signal / group	Direction	Width	Contract
CSR and activity	Input/output	8-bit address; 32-bit write/read data; scalar control	csr_valid/ready accepts csr_write, csr_addr and csr_ wdata; csr_rsp_valid/ready retains csr_rdata and csr_ error. Only one response is outstanding. busy includes active, queued, scrub and reserved-output work, but not merely suspended contexts.

Parameter	Default	Meaning / supported values
MAX_PAYLOAD_ BLOCKS	16	Integer 1..256; 0 and 257 receive explicit parameter diagnostics. Each context retains 16 times this value in payload bytes: 256 bytes by default and 4096 maximum. Four contexts retain 1024 default payload bytes plus 4096 destination-mapping bytes. Context records total 640 bytes and keys total 128 bytes. These are logical capacities, not physical area.

3 Behavior and corner cases

3.1 Handshake and ordering

Transfers require valid and ready immediately before a rising edge; effects appear afterward. Stalled producers hold all fields stable. Newly produced responses cannot retire on their creation edge. Retirement releases dependent ownership afterward. Simultaneous queue enqueue/dequeue executes the old head and preserves FIFO order.

3.2 Descriptor and status encoding

The 64-byte descriptor places byte i at descriptor[8*i+:8]. LE32 fields at offsets 0/4/8/12/16/20/24 hold control/AAD address/source/destination/status/AAD count/payload count. Bytes28..39 hold chronological IV, 40..55 expected tag, 56 cookie and 57..63 zero. Control contains mode1:0 (ECB0, CTR1, GCM2), decrypt2, context4:3, slot6:5, first7 and last8; other bits are zero. Status occupies 32 bytes: LE32 code and cumulative payload count, cookie8, context9, final10, zero11..15 and chronological tag16..31. Tags are zero except successful final GCM.

3.3 Cryptographic arithmetic

Crypto byte0 occupies bits127:120. ECB supports encryption/decryption, no AAD and whole-block lengths, including empty work. CTR uses AES encryption in either direction, begins at IV||1 and increments only the low32-bit big-endian counter. GCM uses J0=IV||1, first payload counter IV||2, H=AES_K(0), ciphertext GHASH and aggregate64-bit big-endian bit lengths. All128 tag bits participate in verification. Counter reuse and length overflow are rejected.

3.4 Session continuity

ECB/CTR require first and last together. GCM supports first/middle/last and empty fragments at arbitrary byte boundaries. Contexts preserve binding, mode, direction, origin, IV, GHASH accumulator, H, mask, counter, lengths, pending hash bytes, unused keystream and retained count. Padding occurs at the AAD-to-payload transition and finalization, not each fragment. Nonempty AAD after payload begins is invalid. Invalid continuations preserve saved state; errors after consumption scrub the affected session.

3.5 Scheduling and authenticated retirement

Idle precedence is accepted key operation, high DMA, normal DMA, direct task, legacy GCM, then raw ECB. Active descriptors are nonpreemptive. Nonfinal state commits before completion. All enhanced output is retained until finalization. Direct completion retires before payload emission, retaining ownership through final output. DMA completion follows acknowledged payload writes and status delivery. Authentication failure releases no payload and performs no payload write.

3.6 Memory safety and faults

Descriptor/status addresses require16-byte alignment; accessed ranges cannot wrap32-bit addresses. Exact in-place payload is supported; prohibited overlaps are rejected before payload writes. Zero-length sections generate no section accesses. Byte transactions preserve unaligned destinations using exact strobes. Memory faults return code8 without retry; unknown status addresses are not invented after fetch failure. Status-write faults remain observable through completion. Acknowledged writes cannot be rolled back.

3.7 Keys, CSR errors and zeroization

Locks persist until reset; bound or locked writes fail atomically. Codes0..8 mean OK, invalid descriptor/ register, invalid key, locked/bound, invalid context, capacity/length, stream protocol, authentication failure and memory error; 9..15 are reserved. CSR offsets0x00/04/08/0c/10/14 provide ID0x52415633, busy, valid/ lock masks, live mask, sticky W1C errors and guarded context abort. Unsupported, misaligned, permission-invalid or reserved-bit accesses fail. Abort rejects active/queued/reserved references, including any queued DMA, and scrubs before responding. Reset cancels work without completion and zeroizes storage. Reset dominates clear, clear dominates context write, and new errors dominate simultaneous W1C.

4 Verification methods and measured results

4.1 Independent cryptography and tool environment

Method. Byte-oriented oracle independent of DUT tables/control, anchored by FIPS197 AES and published NIST GCM answers; retained tool-version and runner-preservation checks.

Result. Four AES and ten GCM known-answer comparisons passed. Recorded tools were Python3.12.3, Icarus Verilog/vvp12.0 and Yosys0.33. Fourteen evidence-runner regression checks passed.

Evidence: E021, E023, E1062

4.2 Full seeded cross-feature behavior

Method. Compiled RTL simulation comparing payload bytes/counts/padding, tags, cookies, contexts, codes, memory traffic and absence of extra output, with strict public-output X/Z rejection.

Result. All9/9 capacity-seed combinations passed. For seeds17/89/203 respectively, comparison counts were4886/4861/4822 at capacity1, 5042/5115/5055 at capacity3 and5978/6036/5948 at capacity16. Each run recorded12 bus faults,131 authentication rejections,42 reset checks and a simultaneous queue event. The complete ledger is evidence/mandatory-rerun-001/behavior/receipt.json. Recovery repeated the same matrix, not additional configurations.

Evidence: E020, E022, E1063, E1069

4.3 Capacity boundaries and rollover semantics

Method. Integrated simulations at capacities1/3/16/256, illegal-parameter runs, public unit interfaces and independently checked boundary arithmetic.

Result. All four legal configurations passed; 0 and257 were explicitly rejected. Targeted capacity256 behavior performed18,250 comparisons over716,139 clock edges. Unit checks passed106 comparisons, including80 arithmetic cases and four context banks. Capacity256 coverage was targeted, not the full seeded matrix; arithmetic checks did not transmit enormous messages.

Evidence: E019, E1064, E1070

4.4 Substantive integrated synthesis and zeroization structure

Method. Yosys hierarchy checks, generic synth -noabc, consistency assertions and complete netlist inspection, including required named instances and asynchronous-reset storage connections.

Result. Capacity16 synthesis reported2,200,117 generic primitive cells, zero inferred latches and no unresolved cells; recovery reported the same primitive total. The verification helper's2,200,127 JSON cell entries additionally include ten hierarchical module instances. Checked resettable storage included40,960 top-level payload/mapping bits,5120 context bits and1024 key bits. These are structural results, not physical area or a noninterference proof.

Evidence: E024, E1065, E1071

4.5 Complete evidence retention after capture failure

Method. Lossless gzip retention with decompressed size/hash verification and self-inclusive bundle accounting; fresh exclusive outputs preserve earlier failures.

Result. An earlier1,698,395,392-byte bundle exceeded the1,073,741,824-byte capture limit. The repaired current synthesis bundle measured67,204,919 bytes and passed capture. Complete hierarchy/netlist JSON compressed to19,699,202/34,484,933 bytes with verified round trips. Output collisions were resolved through fresh allocations, not evidence deletion.

Evidence: E033, E035, E023, E1065

4.6 Checker sensitivity and portable reproduction

Method. Compile and simulate a disposable XOR-one transformed-byte mutation; separately copy only declared deliverables and perform fresh compilation plus focused cross-feature simulation.

Result. The mutation compiled and was behaviorally rejected by a GCM completion-tag mismatch at capacity1, seed17. Clean reproduction passed the same capacity/seed focused campaign with4886 comparisons, without the reference mount, prebuilt binaries or private state. This was not a fresh full matrix or physical-flow reproduction.

Evidence: E1066, E1067, E1072, E1073

5 Integration and use

Use rtl/*.sv, design/SPEC.md, design/ARCHITECTURE.md and tests/* as deliverable inputs. Compile rivet_a3_pkg.sv before the other eleven RTL files and select rivet_a3_top. Adapted AES primitives are included; the read-only A2 reference is provenance, not a build dependency.

Assert reset low, keep input valids low, release reset and wait for post-release readiness. Load required keyslots and consume key results before dependent work. A suspended context binds its key until final retirement or successful abort.

Keep descriptors and source data stable while owned and prevent writable overlap between outstanding jobs. Supply one response for every accepted memory request, discard pre-reset responses and provide finite back-pressure and response delays.

Use Python3.9 or newer, iverilog, vvp and Yosys on PATH. A focused clean reproduction is invoked from the delivered root by python3 tests/verify.py --suite reproduce --output evidence/recipient-reproduce-001; the destination must not exist. Suite commands are documented in design/SPEC.md and tests/README.md. Allocate fresh destinations rather than reusing recorded output paths.

The synthesis helper handles Yosys0.33 package-import limitations by inserting the verbatim package body into generated inputs; functional simulation uses unchanged package/import RTL. Compressed complete netlists, raw logs and source-bound receipts reside under the recorded suite outputs.

6 Limitations and acceptance boundary

This delivery establishes verified RTL and default-capacity generic synthesis, not placement/routing, technology-specific timing, frequency, power, silicon area, FPGA operation or fabrication sign-off.

The byte-serial memory adapter and shared engines prioritize explicit ownership and correctness. No fixed latency, sustained throughput or fairness bound is claimed; continuous high-priority traffic may starve normal work.

GCM uses96-bit IVs and full128-bit tags. Software must provide unique IVs. Hardware nonce-reuse prevention, side-channel hardening and cryptographic certification are not claimed.

Simulation and structural zeroization checks are bounded, not exhaustive formal proof. Maximum-capacity synthesis and a full capacity256 seeded campaign were not established.

Memory errors cannot undo acknowledged writes. Authentication failure prevents payload writes, but late transport failures can leave partial output. Software must inspect completion status before treating destination contents as successful.

A Source inventory

Production inventory: 12 Verilog/SystemVerilog files; 1548 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/rivet_a3_aes_core.sv	48	45
rtl/rivet_a3_context_store.sv	22	22
rtl/rivet_a3_datapath.sv	86	81
rtl/rivet_a3_ghash.sv	63	61
rtl/rivet_a3_key_schedule.sv	90	86
rtl/rivet_a3_key_table.sv	59	59
rtl/rivet_a3_mixcolumns.sv	24	23
rtl/rivet_a3_pkg.sv	105	97
rtl/rivet_a3_sbox.sv	15	14
rtl/rivet_a3_scheduler_dma.sv	263	263
rtl/rivet_a3_shiftrows.sv	17	16

Production file	Physical lines	Code lines
rtl/rivet_a3_top.sv	783	781

B Verification record index

Recorded check	Outcome	Execution time (s)
environment	passed	0.941
behavior	passed	649.009
boundaries	passed	364.900
synthesis	passed	2391.567
checker_sensitivity	passed	2.660
clean_reproduction	passed	68.002
recovery-002-environment	passed	0.966
recovery-002-behavior	passed	652.063
recovery-002-boundaries	passed	364.392
recovery-002-synthesis	passed	2373.040
recovery-002-checker-sensitivity	passed	2.676
recovery-002-clean-reproduction	passed	67.148

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E019	tests/boundary_units.sv
E020	tests/bridge.sv
E021	tests/oracle.py
E022	tests/scenarios.py
E023	tests/test_runner.py
E024	tests/verify.py
E033	.vulcan/engineering/run-0001/attempt-0006/stdout.log
E035	.vulcan/engineering/run-0001/attempt-0008/stdout.log
E1062	.vulcan/engineering/run-0002/attempt-0003/stdout.log
E1063	.vulcan/engineering/run-0002/attempt-0004/stdout.log
E1064	.vulcan/engineering/run-0002/attempt-0005/stdout.log
E1065	.vulcan/engineering/run-0002/attempt-0006/stdout.log
E1066	.vulcan/engineering/run-0002/attempt-0007/stdout.log
E1067	.vulcan/engineering/run-0002/attempt-0008/stdout.log
E1069	.vulcan/engineering/run-0002/attempt-0010/stdout.log
E1070	.vulcan/engineering/run-0002/attempt-0011/stdout.log
E1071	.vulcan/engineering/run-0002/attempt-0012/stdout.log
E1072	.vulcan/engineering/run-0002/attempt-0013/stdout.log
E1073	.vulcan/engineering/run-0002/attempt-0014/stdout.log

C Document provenance

Engineering was independently accepted for the recorded source and evidence identity. This is not exhaustive proof, hardware deployment or physical signoff. The companion REPORT.md records the exact report approval and source manifest, and links the English/Chinese journeys containing original failures and interventions. A report number identifies a generation attempt, not a new hardware revision.

No developer attribution was supplied; none has been inferred.